KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Deemed to be University U/S 3 of UGC Act, 1956

Software requirements specification

Mobile Banking Application

(Instantpay)

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1.0. Introduction

1.1. Purpose

The purpose of this document is to present a detailed description of the mobile banking application . It will explain the purpose and features of the system, the interfaces of the system, what the system will do, the constraints under which it must operate and how the system will react to external stimuli. It serves as a reference for development , testing and maintenance of the application.

1.2. Scope

The “Instant Pay” is a mobile banking application which encompasses a wide range of functions, features and services designed to provide customers with convenient and secure access to their banking and financial needs through mobile devices.
The scope of the mobile banking system encompasses a wide range of functions , features, and services designed to provide customers with convenient and secure access to their banking and financial needs through mobile devices which includes account management , transaction services , loan services , bill payments , scheduled payments , person to person payment , alerts and notifications for account activity and customer support.

1.3 Definitions, acronyms, and abbreviations

Key terms:
APP : Application
PIN: Personal identification number
OTP : One time password
API : Application programming interface
SDK : software development kit
UI/UX : user interface/ user experience
SRS: software requirements specification
SDD: software design document
DEC: Description
RAT: Rational
DEP: Dependency

FR: Functional Requirements

1.4 References

1.https://cpao.nic.in/pdf/technical_committee_report-final.pdf
2.https://cse.msu.edu/~oldenbu6/cse435/res/srsv2.pdf
3.https://www.cse.chalmers.se/~feldt/courses/reqeng/examples/srs_example_2010_group2.pdf
4.https://m.mu.edu.sa/sites/default/files/content/2019/10/Final%20report.pdf
5. Google
6. Wikipedia

COMMITMENT TO DELIVERING A HIGH-QUALITY UNIVERSITY WEBSITE:

- Our commitment to delivering a high-quality University Website is unwavering. We prioritize performance, security, and reliability to create a digital environment that not only meets but exceeds the expectations of all stakeholders, including students, faculty, and administrators [1].

HOW TESTING CONTRIBUTES TO USER SATISFACTION AND OPERATIONAL EFFICIENCY:

- Thorough testing enhances user satisfaction by ensuring the website's functionality, performance, and security. It also promotes operational efficiency by preventing downtime, data loss, and security breaches.

1.5 Overview

This mobile banking application is designed for smartphones and tablets that allow users to access and manage their banking and financial services on the go. The app provides customers with convenience , security and accessibility. It provides features like user authentication, account overview, transaction history, fund transfer, bill payment, alerts and notifications and customer services.

2 Overall Description

This section will cover general information about the project perspective, functions, and various requirements and constraints. More detailed descriptions of the project requirements and functions will be covered in latter sections.

2.1 Product Perspective

Instamtpay seamlessly integrates with the bank's core banking systems and external financial services, providing a unified user experience. It provides robust security measures like end-to-end encryption and multi-factor authentication. The system supports a wide range of user devices, with different smartphones and its versions and is designed to provide a consistent and user-friendly experience across these platforms. It also manages the flow of data, processing and storing user information securely. The error-handling and recovery procedures are in place to address potential interruptions. Additionally, Instantpay may integrate with third-party services and APIs to offer extended functionality, such as payment processing. The system architecture comprises front-end mobile applications, back-end servers, databases, and cloud-based components. User data protection is a priority, with strict privacy policies, data retention practices, and user consent mechanisms in place. Overall, the product perspective emphasizes the system's role within the broader banking ecosystem and its ability to deliver a secure, accessible, and efficient mobile banking experience to users.

2.2 Product Functions

The end user system will initially need to present a user interface where the user will provide his/her personal details and bank information . This information is stored on the server . The system seamlessly integrates with the bank’s existing systems, including core banking , payment gateways and compliance protocol, ensuring a cohesive and interconnected financial infrastructure. The system maintenance strongly focuses on security adhering to regulatory requirements and employing encryption methods to safeguard sensitive user data.

Instantpay includes the following key functions:

- Account access : User can log into the app and view their account balances and transaction

- Fund transfer: Users can transfer money between accounts to other accounts.
- Bill Payment: It facilitates the payment of bills such as utility bills and credit card payments.
- Loan Payment: It allows borrowers to make scheduled loan payments, which are then processed and recorded by lending institutions.
- Alerts and notifications: It sends real time alerts and notifications for account activity, such as deposit and withdrawals.
- Security features: Implements security measures such as two step authentication and biometric verification to protect user data.
- Scheduled payment: It enables users to automate and manage recurring payment for bills, loans and subscriptions.
- Customer Services: It provides in app customer support options including live chat ,email or phone support.

2.3 User Characteristics

This app is designed for a diverse group of individuals like students, business customers, young adults, travelers, banking professionals and loan recipients. User should have basic knowledge of operating android devices and have an existing bank account.

2.4 Constraints

In the development of Instant Pay , a mobile banking system for a diverse user base, various constraints must be considered . security concerns demand robust authentication mechanisms and encryption to safeguard sensitive financial information . Regulatory compliance is crucial, requiring adherence to data protection laws, financial regulations , and standards . Network connectivity poses challenges for regions with unreliable internet access , so a feature should be available for limited offline functionality .Transaction policies must balance security with user convenience , setting limits and defining protocols . Fraud prevention involves implementing detection mechanisms and user education to mitigate risks . Adequate customer support for remote assistance is essential for

addressing customer queries and issues . Technological advancements must be monitored to keep the system secure , efficient and compliant with evolving standards .

2.5 Assumptions and Dependencies

The system relies on the assumption that users possess smartphones . it also assumes that users are digitally literate and sufficiently aware of security practices , expecting them to safeguard their login credentials and mobile devices. Dependencies include the reliability of third party payment gateways for seamless fund transfer and integration with biometric authentication APIs for secure login processes . The system is dependent on the regulatory compliance of financial institutions . The app's functionality relies on a stable server infrastructure , with dependencies on cloud service providers for data storage and processing .

2.6 Proportioning of Requirements

Immediate priorities of the system are robust security measures and a user friendly interface , while future releases could incorporate advanced biometric authentication , personalized financial insights by AI , integration with emerging technologies like blockchain and cryptocurrency , and integration with wearable devices etc .

3.0 Specific Requirements

1. User Registration and Authentication:
   1.1. User enters personal information, including first and last name.
   1.2. User sets up a secure login by entering a username and password.
   1.3. User verifies identity through a multi-factor authentication process, such as SMS or email confirmation.
   1.4. User profile information is saved securely in the backend database.

2. Transaction Logging and Management:

2.1. User initiates a fund transfer by selecting the recipient and entering the amount.

2.2. Users categorize transactions by assigning labels or tags (e.g., bills, groceries).

2.3. Users attach receipts or notes to transactions for personal record-keeping.

2.4. Transactions are timestamped, and relevant details are stored securely in the transaction history database.

2.5. Users can review and edit transaction details, and mistakes can be corrected with a simple undo feature.

3.Bill Payment and Transaction Management:

3.1. Users initiate bill payments by selecting payees, entering amounts, and scheduling due dates.

3.2. Users set up recurring payments for bills, loans, and subscriptions with customizable frequencies.

3.3. Confirmation notifications are sent to users upon successful payment processing.

3.4. Secure storage of payment history and receipts in the app's transaction log.

4.Loan Management and Payment:

4.1. Users view detailed information about existing loans, including balances, interest rates, and payment schedules.

4.2. User schedules loan payments, selecting preferred payment methods.

4.3. App provides timely reminders for upcoming loan payments and sends confirmations upon successful transactions.

4.4. Users can explore loan refinancing options or additional loan products directly through the app.

5. Scheduled Payments and Reminders:

5.1. Users schedule future transactions for recurring expenses, such as rent or subscriptions.

5.1.1. Recurring transactions can be modified or canceled before the scheduled date.

5.2. Reminders are sent to users in advance of scheduled payments, allowing them to ensure sufficient funds are available.

5.3. Users have the flexibility to set reminders for financial goals, upcoming expenses, or important deadlines.

5.4. App maintains a comprehensive calendar view of scheduled payments and reminders.

6. In-App Chat Customer Support:

6.1. The mobile banking app shall provide users with the ability to initiate real-time chat sessions with customer support representatives.

6.2. The chat support feature must be available 24/7 to address user inquiries, provide assistance with transactions, and resolve issues promptly.

6.3. Users should receive instant notifications within the app for responses from customer support representatives, ensuring timely communication and issue resolution.

3.1 External interface Requirements

This section provides a detailed description of all inputs into and outputs from the system. It also gives a description of the hardware, software and communication interfaces .

3.1.1 User interfaces

The mobile banking system's user interface needs to be easily understood and have a safe login process with biometric authentication choices. Access to important functions like fund transfers, transactions, and account summary should be simple to navigate from the home screen. It should be simple for users to view their transaction history and account balances, and there should be choices for both one-time and ongoing financial transfers. Users should be able to easily monitor, plan, and manage recurring invoices in the bill payment section. They should also receive timely alerts and reminders. The management of authentication techniques must be possible through user-friendly security settings, and responsive design ensures the best possible user

experience on a range of devices. It is imperative to incorporate customer support tools, such in-app chat and FAQ accessibility.

3.1.2 Hardware interfaces

The hardware interface for a mobile banking system encompasses the physical aspects that enable users to interact with the application. It mainly makes use of the functionality of smartphones, authenticating users through buttons, touchscreens, and biometric sensors. In order to ensure smooth operation across diverse hardware configurations, the interface needs to be designed for a range of device characteristics and screen sizes. Furthermore, the consideration of peripheral compatibility, like card readers or fingerprint scanners, can offer flexibility and improve the overall security and usability of the mobile banking system in a variety of hardware setups.

3.1.3 Software interfaces

The software interface of a mobile banking system is the user's digital gateway, featuring an intuitive design for functions like login, transactions, and account management.It easily connects with both iOS and Android operating systems, putting security first with multi-factor authentication and encryption. Real-time updates and seamless touchscreen operations are guaranteed by the interface, and more functionality can be added via API interfaces. Through the optimization of user experience and connectivity, this software bridge guarantees a safe, current, and intuitive interface for efficient mobile banking interactions.

3.1.4 Communications interfaces

The communications interface for a mobile banking system facilitates seamless interactions between the application and external entities. It uses secure communication protocols, to transfer data between the banking servers and the mobile app. Connectivity with third-party services is made possible through API

integrations, which offer functions like real-time updates and fund transfers. Sensitive data is transmitted securely through the interface, which also enables encrypted connection for user authentication.

3.2. Functional Requirements

This section includes the requirements that specify all the fundamental actions of the mobile banking application software.

3.2.1 User class 1 - The user
3.2.1.1 Functional requirement 1.1
ID: FR1
TITLE: Download mobile banking application
DESC: A user should be able to download the mobile application through either an application store or similar service on the mobile phone. The application should be free to download.
RAT: In order for a user to download the mobile banking application.
DEP: None

3.2.1.2 Functional requirement 1.2

ID: FR2
TITLE: Receive Notifications for updates
DESC: The mobile banking app should notify users of any new features, updates, or releases. Customers can choose to update the app manually through the app store or receive automatic updates, ensuring they have access to the latest functionalities.
RAT: In order for a user to stay informed and access new features .
DEP: FR1

3.2.1.3 Functional requirement 1.3

ID: FR3
TITLE: User registration - Mobile application
DESC: Given that a user has downloaded the mobile application, then the user should be able to register through the mobile application. The user must provide first name, last name, password ,account details and email address. The user can choose to

provide a regularly used phone number. Also biometric details are taken for security purposes.
RAT: In order for a user to register on the mobile application.
DEP: FR1

3.2.1.4 Functional requirement 1.4

ID: FR4
TITLE: User log-in - Mobile application
DESC: Given that a user has registered, then the user should be able to log in to the mobile application. The log-in information will be stored on the phone and in the future the user should be logged in automatically.
RAT: In order for a user to register on the mobile application.
DEP: FR1, FR3

3.2.1.5 Functional requirement 1.5

ID: FR5
TITLE: Retrieve password
DESC: Given that a user has registered, then the user should be able to retrieve his/her password by email or by otp sent on registered number.
RAT: In order for a user to retrieve his/her password.
DEP: FR1

3.2.1.6 Functional requirement 1.6

ID: FR6
TITLE: Account Management
DESC:Users should be able to view their account balance and recent transactions and should be able to link multiple bank accounts to their mobile banking app.
They should have the ability to transfer funds between their linked accounts.
The system should support various types of accounts (savings, current etc.).
RAT: For user to clearly see how their accounts to be used
DEP: FR3

3.2.1.7 Functional requirement 1.7

ID: FR7

TITLE:Fund Transfers
DESC:Users should be able to initiate transfers to other bank accounts, both within and outside the bank.Transfer options may include one-time transfers, recurring transfers, and scheduled transfers.The system should provide confirmation and notifications for successful and failed transactions.
RAT: For transfering funds easily without hustle.
DEP:FR3,FR6

3.2.1.8 Functional requirement 1.8

ID: FR8
TITLE:Bill Payments
DESC:Users should be able to pay bills such as utilities, credit cards, and loans.
The system should allow users to set up recurring bill payments. They should receive notifications and confirmations for bill payments.
RAT: For bill payments
DEP:FR3,FR6

3.2.1.9 Functional requirement 1.9

ID: FR9
TITLE:Transaction History
DESC:Users should be able to view and search their transaction history with filtering options.The transaction history should include details like date, time, payee, amount, and transaction status.
RAT: For showing transaction history and details of the transaction.
DEP:FR1,FR6,FR7,FR8

3.2.1.10 Functional requirement 1.10

ID: FR10
TITLE:Alerts and notifications
DESC:Users should have the option to set up alerts and notifications for account activity, low balances, large transactions, etc.Notifications should be delivered through push notifications and/or email and text messages on registered phone number.
RAT: For receiving alerts and notifications for every actions.
DEP:FR1,FR2,FR3,FR4,FR5,FR6,FR7,FR8

3.2.1.11 Functional requirement 1.11

ID: FR11
TITLE:Security and Fraud Prediction
DESC:The application should implement security features such as encryption, biometric authentication, and two-step authentication.The system should monitor and flag suspicious transactions and account activity for review.
RAT: For Security and fraud protection
DESC:FR1,FR2,FR4,FR5,FR9,FR10

3.2.1.12 Functional requirement 1.12

ID: FR12
TITLE:Customer Support and Help center
DESC:The application should offer customer support through chat, email, or phone.A help center or FAQ section should be available to assist users with common issues and inquiries.
RAT: For solving queries which occur to the users and help with problems.
DESC:FR1,FR2,FR3,FR4,FR5,FR6,FR7,FR8,FR9,FR10,FR11

3.2.1.13 Functional requirement 1.13

ID: FR13
TITLE:Profile Management
DESC:Users should be able to update their personal information, contact details, and account preferences.The system should allow users to change their login credentials.
RAT: For profile maintenance and updating information
DESC:FR1,FR2

3.2.2 User class 2 - The Administrator

3.2.2.1 Functional requirement 2.1

ID : FR 14
TITLE: Admin User Authentication and Access Control

Desc: Admin users should have a secure authentication process, including password and multi-factor authentication, to access administrator functionalities.
RAT: In order for an administrator to securely access and manage the system.
DEP: None

3.2.2.2 Functional requirement 2.2

ID : FR 15
Title: User Management - Admin
Desc: Admins can view and manage user accounts, including creating new accounts, modifying user details, and deactivating accounts if necessary.
RAT: In order for administrators to manage user accounts efficiently.
DEP: FR14

3.2.2.3 Functional requirement 2.3

ID : FR 16
Title: Transaction Monitoring and Reporting
Desc: Admins should have access to real-time transaction monitoring, with the ability to generate reports on user activities and transaction histories.
RAT: In order for administrators to monitor and analyze system transactions.
DEP: FR14

3.2.2.4 Functional requirement 2.4

ID : FR 17
Title: System Configuration and Settings
Desc: Admins can configure system settings, including security parameters, notification preferences, and other administrative configurations.
RAT: In order for administrators to customize and optimize system functionalities.
DEP: FR14

3.2.2.5 Functional requirement 2.5

ID : FR 18
Title: Security Incident Response

Desc: Admins should receive immediate notifications for security incidents and have the authority to respond promptly, including the ability to lock accounts or initiate additional security measures.
RAT: In order for administrators to address and mitigate security incidents swiftly.
DEP: FR14, FR16

3.2.2.6 Functional requirement 2.6

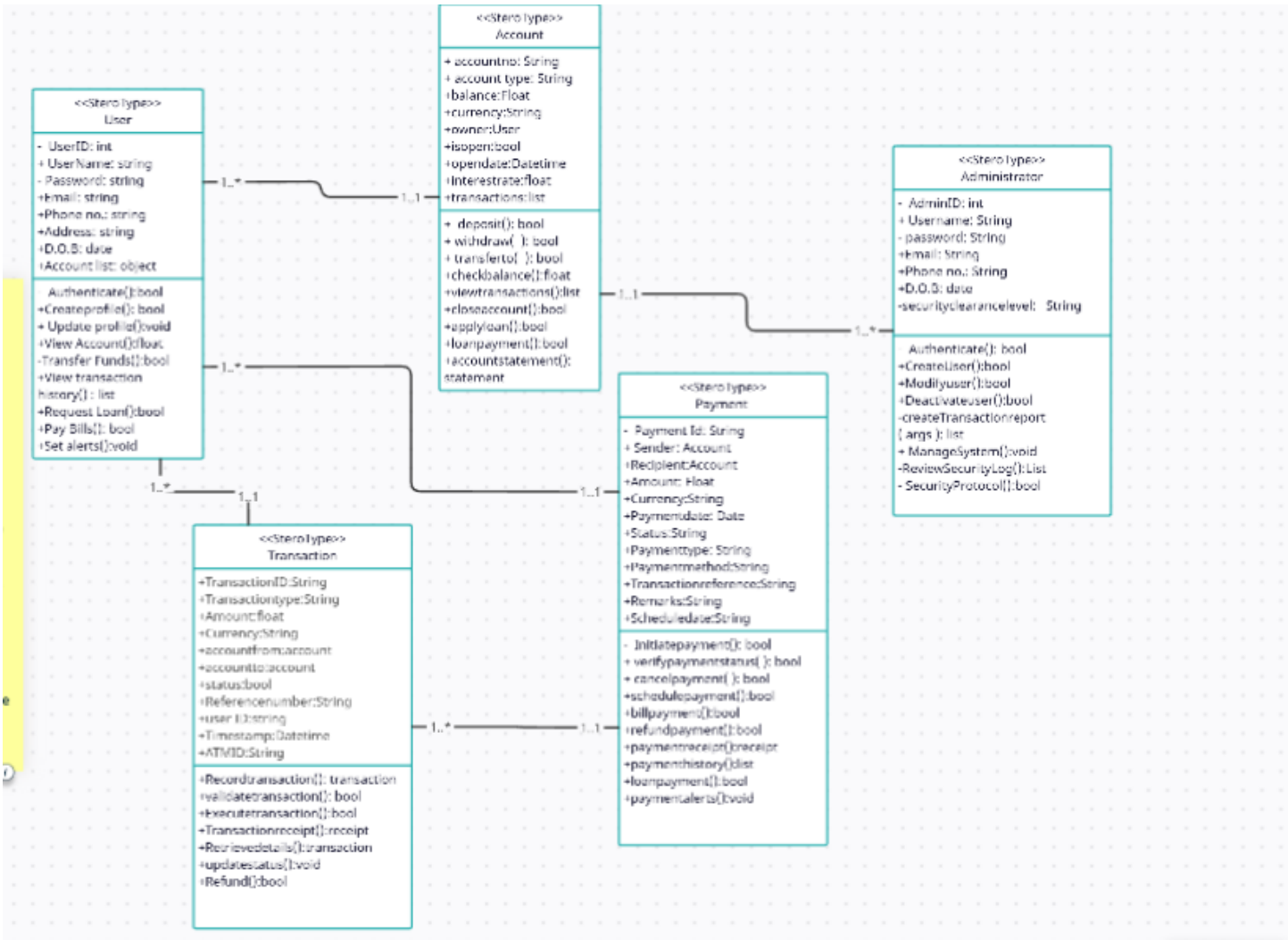
ID : FR 19
Title: User Support and Issue Resolution
Desc: Admins have access to a comprehensive support system, enabling them to assist users, resolve issues, and provide necessary support.
RAT: In order for administrators to efficiently support and address user concerns.
DEP: FR14

4 Modeling Requirements

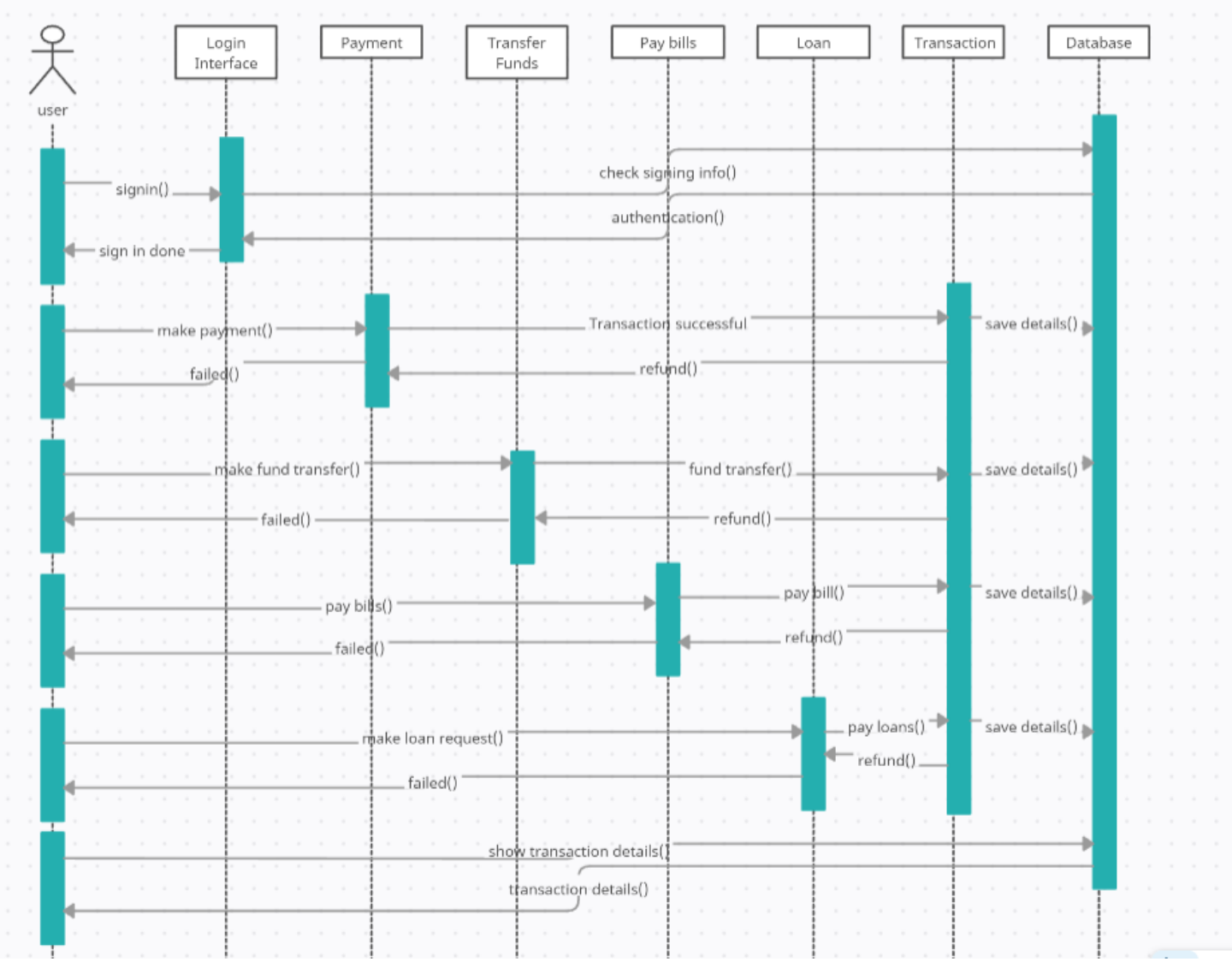
This section provides use case, class, sequence, and state diagrams that address the specifications of the requirements defined in section 3.

1. Use case diagrams : illustrate possible situations where the user interacts with the system.

2. Class Diagram:



3. Sequential diagrams



5.1 Non-Functional Requirements

- **Performance**
  When sending a help request an alert will be sent to all registrants in the program.
- **Usability**
  Ease of use and clarity of the program.
- **Locating help**
  Send the user a location to access if needed.
- **Scalability**

The program is scalable and modifiable by the Administrator.

- **Interoperability**
  The program is based on helping Users to each other.
- **Reliability**
  The program is reliable during use and browsing inside.
- **Maintainability**
  The program is maintenanceable when there is any problem.
- **Serviceability**
  User can communicate with the administrator when there is any inquiry or suggestion.
- **Security**
  The program is secured from hacking and viruses.
- **Regulatory**
  Working within the program is regular and easy to use.
- **Manageability**
  There is an administrator who manages and supervises the program.

6.1 Test Classes And Overall Test Conditions

In the next sections of the document, many test cases are given a place for checking the software in terms of different perspectives.

6.2 Details For System Plan

6.2.1 Features To Be Tested

Projects functional properties will be tested. While testing environment variables and specific input parameters will be used to see the software's behavior and the related errors.

6.2.3 Approach

Blackbox testing which examines the functionality of an application without peering into its internal structures or workings is determined to be used based on our knowledge of internal operation and implementation of Instantpay.

6.2.4 Item Pass / Fail Criteria

For each test there are input- -output pairs which must be satisfied by related test cases. The test cases which result in unexpected outputs shall be considered as fail. This approach shall be applied only to test cases with black box method.

6.2.5 Test Deliverables

This document contains test procedure specifications that explain the steps for executing all test cases given in the following sections and the results of each test case. It is prepared based on IEEE Standard for Software and System Test Documentation.

6.3. Test Management

6.3.1 Planned Activities and Tasks; Test Progression

The test process will start with analysis and inspection of the SRS in order to understand the requirements and design cases and code in order to find the error prone parts of the code for testing purposes. Then, necessary test cases will be determined and for better understanding, cases will be divided into groups according to their objectives. After that, the expected outputs for each test case will be decided and described. Finally, inputs for each test case will be determined.

In the final step, the results of each test will be gathered in a test results table. This will enable us to see what should be done in further steps of the software’s development. The testing of this project will be done manually except performance testing.

6.4 Test Case Details

6.4.1 Introduction

This section provides the detailed explanation for each test case accompanied by the its inputs, outcomes, environmental and procedural requirements along with the dependencies among test cases. This section includes the information for all test cases we run on the project.

6.4.2 Test Cases

Test case identifier: Test1
Objective :Logging in
Inputs : username, password
Outcome: User is logged in

Test case Identifier : Test 2
Objective :Logging out
Inputs : quit command
Outcome: User is logged out

Test case Identifier : Test 3
Objective : transfer funds
Inputs : transaction id , amount , accountto,accountfrom,currency
Outcome: transaction successful

Test case Identifier : Test 4
Objective : pay bills
Inputs : transaction id , amount , account to,account for,currency
Outcome: payment successful successful

Test case Identifier : Test 5
Objective : pay loan
Inputs : balance , amount , accountto,accountfrom,interest

Outcome: payment successful

6.5.1 Overview Of Test Results

In this section of the document, test cases that are explained in the previous section are concluded with their results according to pass/fail criteria that is also defined in the document.

Results of the all test cases are given in following part.

6 5.2 Detailed Test Results

Test1: PASS
Test2 :PASS
Test3 :PASS
Test4: PASS
Test5 : PASS

6.5.3 Rationale For Decisions

Test cases were formed in order to see the behavior of the code in prone parameters and inputs. The test cases were tested number of times and behavior of the system was observed using our predefined evaluation metrics.

6 5.4 . Conclusions And recommendations

In the testing period ,5 different test cases are executed. All of the test are about functionality testing about the project.

7 Reference

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